Horizon-dependent Trade-offs and Undercoverage in Probabilistic Movement Forecasting

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Abstract

A movement forecast can improve its distributional score without locating the target more accurately or providing reliable uncertainty regions. We examine these distinctions in an existing evaluation of fitted stochastic movement predictors, using 46 retrospectively selected recording-hash blocks, five simulation seeds and 512 particles per ensemble. At nominal 1 and 5 minutes, deterministic constant-speed extrapolation has lower energy score (ES) than the reference stochastic predictor Full; at 15 and 30 minutes the ordering reverses. Averaging the four targets gives ES 477.85 m for Full and 824.73 m for inertia, concealing that reversal. Within the stochastic procedures, a 300 s observation-interval variant reduces average ES by 23.40 m but increases final mean-position error by 9.64 m relative to its named control. Full's nominal 90% prediction disk covers only 46.52% of 30-minute targets. These are conditional descriptive findings: participant independence and physical clock provenance remain unverified, and the cohort was selected using future-route map support. Required forecast failures prevent terrain-family inference. The study shows why horizon-specific score, mean-position error and region coverage must be reported together; it does not establish stochastic-model superiority over a simple probabilistic baseline or operational reliability.

1 Introduction

Movement prediction is often summarized by a single accuracy measure, yet three distinct tasks are involved: assigning a distribution to future positions, locating a representative position, and describing a region that contains the realized target at its stated frequency. A gain on one task need not imply a gain on the others. This distinction matters when a probabilistic forecast is interpreted as a reliable search region.

Continuous-time correlated random walks model movement persistence and irregular observations [1]; switching state-space models represent latent regimes [2]. Step-selection models link movement to environmental selection [3, 4]. Learned trajectory predictors [5, 6], neural SDE generators [7] and interpretable stochastic system identification [8] offer additional modelling approaches. These works provide context, not performance baselines executed here. Our subject is an existing, limited collection of fitted stochastic predictors, not a new SDE theory or a state-of-the-art benchmark.

Proper scoring rules evaluate predictive distributions against realized targets [9, 10]; they do not certify coverage of a particular prediction region. We therefore ask how distributional score, error of the predictive mean and marginal disk coverage change across four horizons. The empirical contribution has three parts: a short/long-horizon reversal hidden by the average score, an example of score improvement with worse mean-position error, and severe long-horizon undercoverage despite substantial region area.

This focus is an interpretation of existing results, not a newly registered hypothesis test. The original questions remain explicit: RQ1 concerns method-component comparisons, and RQ2 concerns terrain increments beyond a history baseline. RQ1 yields conditional comparisons of implemented procedures; RQ2 remains unavailable under its complete-family rule. Neither is redefined to turn an inconclusive or unavailable result into a positive finding.

2 Data and predictive procedures

2.1 Evaluation population and clock

The task is single-recording position forecasting from a visible trajectory prefix. The retained data pipeline identifies its source as a historical OpenStreetMap GPX harvest, not the separate GeoLife cohort used in earlier work. User identities, transport-mode labels and original GPX segment boundaries are not preserved; these records cannot be treated as verified pedestrian participants. R1-derived segments supply midpoint history/target windows. The frozen recording-hash assignment separates development and evaluation: qualified development contains 404 outer-training windows (328 training and 76 adaptation) and 81 validation windows. Terrain fitting uses one first-future transition per outer-training window. Exact identity intersections across splits are zero, which does not establish participant or near-route independence. The scientific supplement details this assignment.

The released final-evaluation split contains 12,370 windows from 1,094 recording-hash blocks. Temporal eligibility retains 127 windows/89 blocks, joint feature validity retains 106/73, and the fixed primary-origin rule selects 46/46 (Figure 1). A block identifies a saved recording, not a verified independent participant. The selected cohort spans 12 countries and 27 source labels; 33/46 blocks are from Great Britain, Belgium or Luxembourg. It is not a representative population sample.

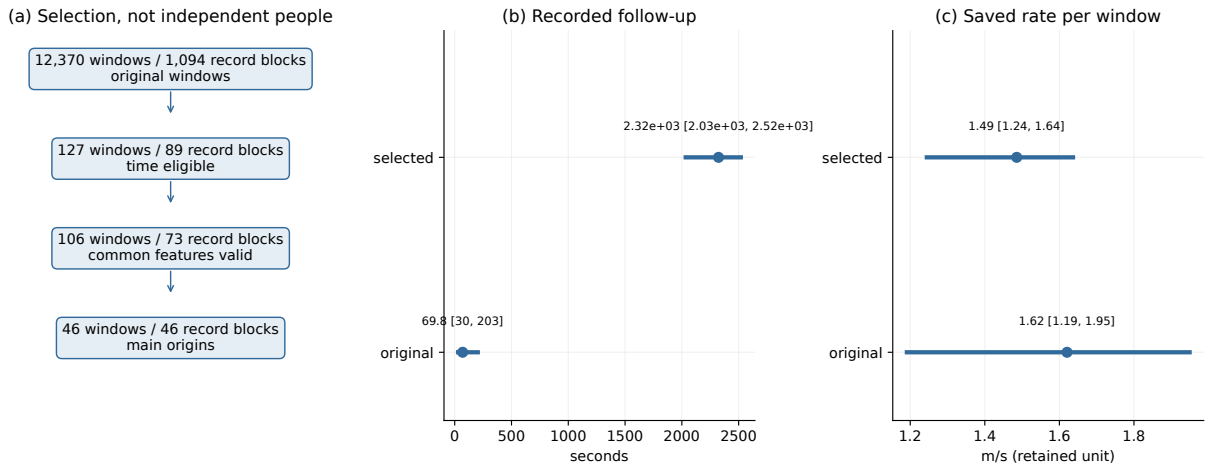


Figure 1: Evaluation selection and population descriptors. Counts distinguish windows from recording blocks; duration and speed panels show saved medians and interquartile ranges. Eligibility includes map validity along the observed future route, which is unavailable at deployment from the prefix alone. Selection exclusions are not forecast failures.

Future observations are used to select this cohort and to score forecasts, but are not inputs to prediction: online map queries use simulated positions. This distinction prevents direct target leakage during rollout without removing the retrospective selection restriction. The evaluation

data had previously been inspected; the full influence of that inspection on design choices is not recoverable. This is not a fresh blinded holdout.

Targets are the actual observations matched to nominal 1, 5, 15 and 30 minutes, with elapsed-time ranges 49–67, 294–305, 889–910 and 1785–1814 saved seconds. No target is interpolated to an exact nominal time. Physical acquisition-time and UTC provenance are unverified. An absolute offset affects the calendar input; a relative-scale distortion also affects rates, diffusion units and the physical interpretation of the horizons. All minute labels below refer to the saved clock.

2.2 Reference stochastic predictor and comparators

In the descriptive reference comparisons, Full denotes Full01, the ordinary three-mode fitted reference, not the all-terrain configuration. In saved local east/north coordinates $X = (X_e, X_n)^\top$ (metres), it uses

$$dX_t = \Theta_{m_t}^\top \psi(X_t, t)\, dt + L_{m_t}\, dW_t, \quad \psi = (1, X_e, X_n, S)^\top, \quad L_m L_m^\top = Q_m = \tau R_m. \tag{1}$$

The first visible position anchors the coordinate frame; forecasting starts at the last permitted prefix position. S is a solar/calendar proxy computed from the retained absolute tick [15], not a validated daylight mechanism. R_m is a centred fitted displacement-rate residual covariance; its conversion uses observation interval τ, not integration step h. Each particle draws a mode from saved weights π at the origin and segment-fixed Full retains that mode. Given frame, position, clock and random stream, ordinary Full does not read initial speed or recent heading. For the 46 primary origins, the visible input retains the last three observed positions and their strictly increasing saved times. The origin is the last point. Inertia extrapolates $\widehat{x}^{\mathrm{I}}(u) = x_0 + u\widehat{v}_0$, where $\widehat{v}_0$ is the secant from the first to last of those three points, divided by their saved elapsed time; no future position or future speed is used. These spans range from 2 to 61 saved seconds (median 7). Ordinary Full and inertia thus receive the same permitted prefix but use its movement information differently.

Drift is fitted by ridge regression. Full selects a finite covariance-scale grid with drift blending fixed at zero. The selection proxy is a one-step rate-residual Gaussian using the unweighted average mode covariance, whereas evaluation scores a multi-mode, multi-step position distribution. The GMM variant groups training residuals once by $\arg\max(r_e, r_n, -r_e - r_n)$ but uses segment-heading labels for validation components: these labels are not aligned. Independently heading-sorted batches are also mixed by mode index in the transfer procedures [12]. These are implementation limitations, not evidence that the corresponding modelling ideas fail.

Evaluation uses $N = 512$ particles, five fixed seeds and maximum integration step $h = 5$ saved seconds. Full uses conditional Gaussian moment propagation, evaluating nonlinear conditions at the conditional mean; this is not an exact solution of a globally nonlinear SDE [11]. Four targets come from a continuing joint path, not independently sampled marginals. Observation-interval variants change fitting data, $Q = \tau R$ and update cadence together while retaining the integration cap. In particular, dt300 and dt600 mean observation intervals of 300 and 600 saved seconds, not integration steps. Identical-parameter aliases and multi-factor changes make the 28 saved identities procedures rather than 28 independent algorithms. All 21 registered primary contrasts retain their named Full controls.

2.3 Terrain branch and scope of attribution

The separate terrain branch augments a fitted three-position history drift with independently fitted residual conditioners and diffusion for ten configurations. It includes retained roads, rivers, land cover and surface orientation, with interactions. Model-source declarations identify WorldCover

2021 v200 [16], Overture transportation 2026-08-19.0 [23, 17], HydroRIVERS v1.0 [18, 19], and retained Mapzen Skadi / Copernicus GLO-30 rules [20, 21, 22]. These product declarations do not certify tile/track contemporaneity. Road/river direction points to the nearest feature, not along its tangent or flow; the retained roads are not the complete pedestrian network.

Constant validity masks, penalised intercepts, unstandardised features and refitted diffusion make effective regularisation differ among configurations. Thus even a complete comparison would concern configuration increments, not isolated terrain information. The scientific supplement gives full equations, input geometry, excluded road classes, fitting rules and procedure-specific controls. The terrain research question remains separate from ordinary Full.

3 Evaluation measures and inference

For particles $x_1, \ldots, x_N$ and target y at one matched horizon, the execution-bound finite-ensemble energy score is

$$\mathrm{ES}_N = \frac{1}{N}\sum_i \|x_i - y\| - \frac{1}{2N(N-1)}\sum_{i\neq j}\|x_i - x_j\|, \qquad N \geq 2. \tag{2}$$

The second sum uses ordered pairs. For the deterministic inertial reference, ES is directly $\|x - y\|$. ES is measured in metres and lower is better; weighted ES is the equally weighted average over the four targets. Mean-position error is $e_j = \|\bar{x}_j - y_j\|$, with $\mathrm{ADE} = \frac{1}{4}\sum_j e_j$ and $\mathrm{FDE} = e_4$. FDE is neither best-particle nor average-particle error, and ADE is not a dense-path integral. ES includes interparticle spread and cannot be converted into error of the predictive mean.

A nominal level-q disk is centred at the ensemble mean, with radius the $\lceil qN \rceil$-th ordered distance of particles from that mean. Coverage records whether the true target lies inside; area is πr^2. Mean area is $\mathrm{E}[\pi r^2]$, not $\pi \mathrm{E}[r]^2$. These are horizon-wise marginal regions, not joint path regions. Enclosing about 90% of simulated particles does not guarantee 90% coverage of real targets. For multimodal clouds, the mean may lie in a low-density gap and a disk may include inaccessible terrain; it is not a highest-density or route-constrained region.

For each recording block, the five seed values are averaged first; blocks then receive equal weight. There are 46 targets per horizon, not 230 independent targets. Six known-velocity and six point-only cases reuse primary records, giving 58 origin–mode cases rather than 58 independent blocks. Original family-wise simultaneous bootstrap-t intervals use 2,000 draws per stream, with shared block indices within a family; Holm-adjusted zero tests have a separate role [13, 14]. Intervals condition on the saved fits and unverified block independence / exchangeability; seeds represent simulation variation, not training or participant replication.

Deltas are candidate minus named control. The registered practical margin is $\delta = 41.100259$ m, 5% of validation inertial ES, not a validated operational utility threshold. Practical benefit/harm requires the entire original interval beyond $-\delta/+\delta$ and the original statistical, seed and qualification gates. Crossing zero is not equivalence. Practical equivalence, inconclusive information and unavailable evidence are distinct. The scientific supplement retains precise endpoints and finite bootstrap-tail limitations. The descriptive horizon and coverage summaries below are not new superiority or calibration hypothesis tests.

4 Results

4.1 Average score conceals a horizon reversal

Full's weighted ES is 477.85 m versus 824.73 m for inertia; the saved difference is -346.87 m, computed before rounding. At 1 minute, Full/inertia ES is 50.57/31.17 m; at 5 minutes it is 243.23/237.93 m. Thus inertia has lower ES at 1 and 5 minutes and Full at 15 and 30 minutes (Figure 2). The same figure separates ES from mean-position error. Four discrete targets do not identify an exact crossover time. The deterministic reference is a valid point-mass distribution

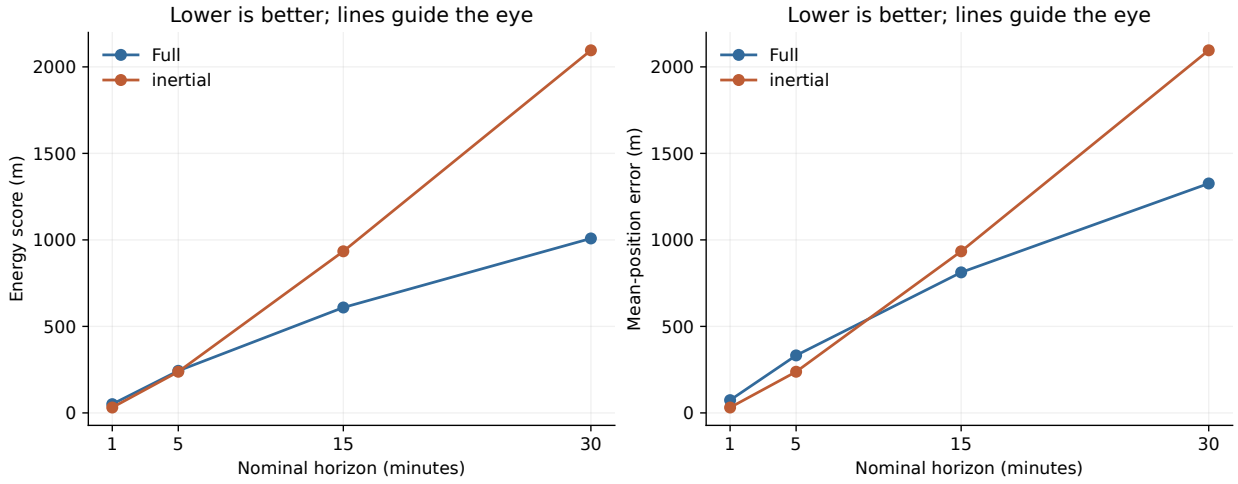


Figure 2: Full versus deterministic inertia at four nominal saved-clock horizons: ES (left) and predictive-mean error (right). Each point averages five seeds within each of 46 primary blocks, then weights blocks equally; inertia is deterministic. Lower is better. Connecting lines guide reading, not interpolation or evidence for a precise crossover.

Table 1: Saved predictive-mean errors (m), using the original matched targets. These are not ES or mean particle errors.

Model	1 min	5 min	15 min	30 min
Full	73.77	332.31	812.26	1326.09
GMM	71.92	323.79	792.99	1299.11
dt300	75.88	340.10	825.76	1335.73
Inertial	31.17	237.93	934.03	2095.78

for ES. It does not establish that Full's stochastic complexity is necessary relative to a simple probabilistic constant-velocity or correlated-random-walk model.

4.2 A better distribution score need not locate the mean better

The 300 s observation-interval procedure illustrates the distinction. Relative to its named Full01 control, weighted ES changes by −23.40 m but FDE by +9.64 m. Its mean-position error is also larger at each of the four saved targets (Table 1). ES accounts both for distances to the target and

for spread between particles; it cannot be converted into mean-position error. The observation-interval comparison changes training exposure and diffusion binding as well as cadence, so this contrast does not isolate a numerical integration effect.

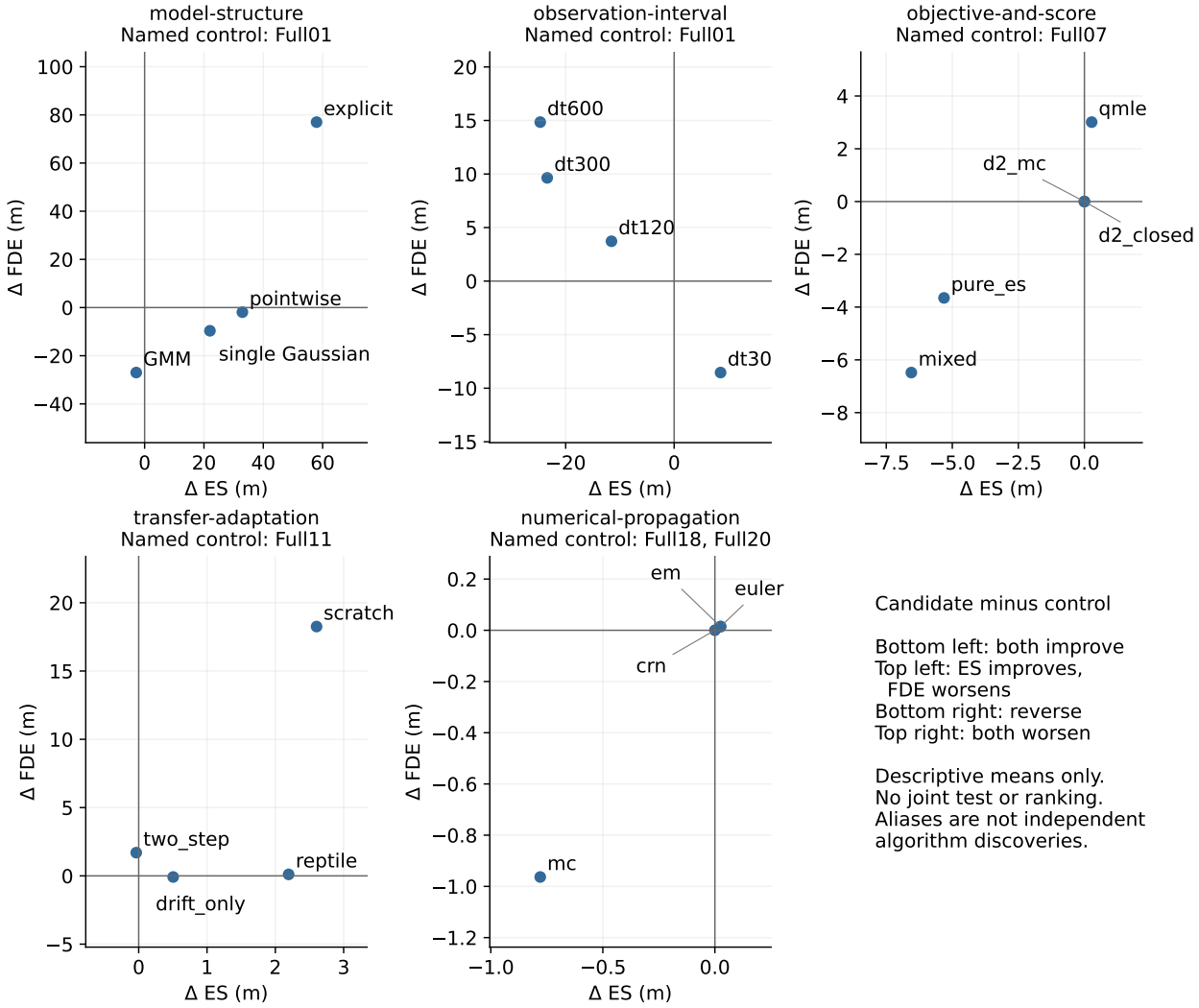


Figure 3: Distribution-score versus endpoint-error trade-offs. All 21 comparisons (42 displayed scalar differences) retain five registered families and named controls. Each point averages five simulation seeds within each of 46 paired blocks, then weights blocks equally; ES first averages four targets while FDE uses only the last. Negative is favourable on both axes; the upper-left quadrant improves ES but worsens FDE. Facets have different scales. These means have no joint confidence region and do not establish a Pareto or global algorithm ranking.

The dt300 ES simultaneous interval is $[-48.27, -2.47]$ m. Excluding zero does not satisfy the registered practical-benefit criterion: the whole interval would have to be below -41.100259 m, with the other gates met. The point thus communicates a metric conflict, not a practically superior method. All differences are calculated from unrounded source values.

4.3 Long-horizon disks substantially undercover

Full's nominal 90% disk coverage is 95.65%, 37.83%, 35.22% and 46.52% at the four horizons. At one minute the nominal 50% disk covers 33.04%, while at thirty minutes even the nominal 95% disk covers only 52.17%. The departure therefore depends on horizon and probability level; no radius rescaling has been tested. Across all 28 identities, thirty-minute 90% coverage ranges from 21.74% to 79.13%. The dt300 disk is about twice Full's mean area, yet its inclusion rate remains below the

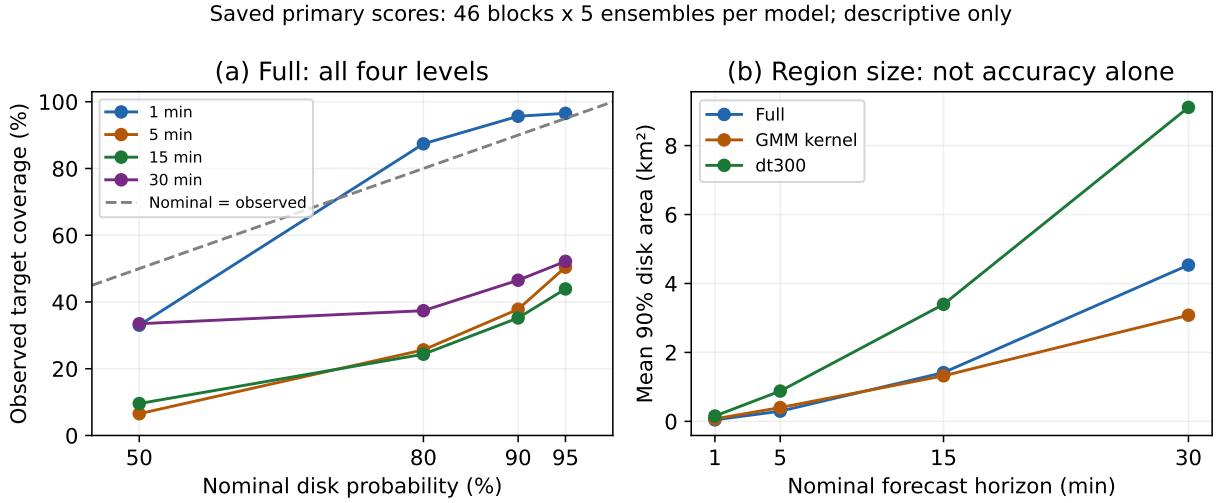


Figure 4: Calibration and size from the saved 46-block, five-seed summaries. Left: 16 Full coverage values (four probability levels at four horizons); closeness to the diagonal, not the highest curve, is the target. Right: 12 mean areas (three configurations at four horizons), to be read with coverage rather than ranked alone. Connecting lines guide reading; they do not add intermediate horizons or a calibration test.

Table 2: Thirty-minute nominal 90% disks: three configurations and nine saved summary values. Each configuration uses 46 blocks and five ensembles per block. Radius and area are averaged separately; the radius of the mean area is not the mean radius.

Model	Coverage (%)	Mean area (km^2)	Mean radius (m)
Full01	46.52	4.534	1183.6
GMM	45.22	3.079	989.5
dt300	64.35	9.109	1692.8

nominal 90%. A larger region can include more targets while its centre is less accurate. Conversely, GMM's smaller region does not by itself establish better uncertainty prediction. Coverage averages the same real target across five simulated disks in each block; there are 46 targets per horizon, not 230 independent targets. At block level this five-seed inclusion proportion takes only 0, 20, 40, 60, 80 or 100%. The 552 points in Supplement Figure 16 can overlap at those six levels; neither a block median of 100% nor a dense-looking cluster is a conditional calibration certificate. These observations support reporting size and inclusion together while retaining the complete block summaries. The 600 s procedure also has a larger thirty-minute mean area (12.609 km^2) and higher coverage (79.13%) than Full, but its FDE is 1340.94 m. This is another descriptive size–coverage

trade-off, not evidence of calibrated uncertainty or improved mean location. No radius or parameter is retuned after inspection of this cohort.

4.4 Limits of component and terrain conclusions

Of the 21 registered primary method contrasts, 14 have practically equivalent and seven inconclusive finite-algorithm states; no original simultaneous interval lies wholly beyond the practical-benefit or harm boundary. All resolution-invariant states remain inconclusive. These states qualify the implemented finite procedures, not general algorithms. For example, all five GMM seed effects are negative but its block interval is $[-21.22, 16.33]$ m. Seed-direction agreement does not resolve uncertainty across recording blocks. Complete effects, seeds and qualification reasons are retained in Supplement Table 25 and the accompanying qualification discussion.

Three required primary terrain forecasts failed: all-terrain once and LIO-river/LIO-road once each. LOO means leave-one-group-out and LIO means leave-one-group-in. The overall/LOO grid has 1379/1380 successes and LIO has 1148/1150, with no pending forecasts; the grids share baselines and are not additive. Under the original complete-family rule, both whole families are unavailable. Successful subsets or individual map cases cannot replace that rule. This is not a positive or null terrain result. Two further failures affect auxiliary known-velocity method families, not the complete primary method families. Failure identities and logging limits are in the audit record; missing exception stacks prevent attributing a specific technical cause.

5 Discussion

The three findings address different decisions. The horizon reversal warns against choosing a predictor from its average score alone. The dt300 contrast shows that a distribution-score gain does not imply a more accurate representative location. The coverage results show that a model-internal particle quantile is not an empirical reliability guarantee. Together they support a reporting rule: pair horizon-specific distribution scores with mean-position errors and the coverage and size of any proposed region. This is a diagnostic implication, not a validated operational policy.

The calibration proxy differs from the evaluated predictive distribution, and GMM training/validation component labels differ. Both are plausible contributors to poor uncertainty behaviour, but these results do not identify causes or validate a correction. Likewise, wider regions can increase inclusion while retaining an inaccurate centre; coverage alone would reward sufficiently uninformative regions. The metric conflict is therefore substantive, not a reason to discard an unfavourable metric.

The scope is deliberately conditional. Retrospective future-route selection, previous evaluation exposure, uncertain recording independence and unverified clock provenance limit population and physical-time interpretation. Fixed fits and seeds omit training uncertainty. Parameter aliases, unmatched mode labels, joint procedure changes and representation-dependent penalties limit component attribution. The missing matched probabilistic baseline precludes a claim that complex stochastic structure is necessary. None of these limitations is removed by reproducing saved numerical summaries.

Future work should compare a task-matched probabilistic motion baseline, align mode labels before transfer and calibration, and evaluate calibration of the actual multi-step distribution. Terrain comparisons require complete families and matched effective penalties. Independent route grouping and clock-source verification are needed for broader inference. These are unexecuted studies, not results or editorial repairs supplied here.

6 Conclusion

In this retrospective cohort, a favourable average energy score hides a short/long-horizon reversal, a score gain can accompany worse endpoint location, and nominal 90% disks can cover fewer than half of long-horizon targets. Joint horizon-wise reporting exposes these differences. RQ1 supports bounded finite-procedure comparisons, while RQ2 remains unavailable; neither establishes general algorithm superiority or terrain efficacy.

Data and reproducibility. The accompanying scientific supplement contains full specifications and all 50 tables; the separate audit record contains execution and revision history. The public repository provides saved aggregate evidence and reproduction code; private trajectories and local case maps are not redistributed. Repository: `https://github.com/HuAzurestar/learnable-sde-paper-to-predict-movement`. The public argument does not rely on those private case illustrations. Trajectory research-use permissions and the applicable ethics-review / consent basis require author verification; public availability of a source or map product is not taken as evidence of that authorization. This manuscript revision changes no fit, forecast, score or inference rule. Author identities, affiliations and contribution/conflict declarations require owner confirmation before submission and are not inferred from repository accounts.

References

[1] D. S. Johnson, J. M. London, M.-A. Lea and J. W. Durban (2008). Continuous-time correlated random walk model for animal telemetry data. *Ecology*, 89(5), 1208–1215. `https://doi.org/10.1890/07-1032.1`.

[2] Z. Ghahramani and G. E. Hinton (2000). Variational learning for switching state-space models. *Neural Computation*, 12(4), 831–864. `https://doi.org/10.1162/089976600300015619`.

[3] T. Avgar, J. R. Potts, M. A. Lewis and M. S. Boyce (2016). Integrated step selection analysis: bridging the gap between resource selection and animal movement. *Methods in Ecology and Evolution*, 7(5), 619–630. `https://doi.org/10.1111/2041-210X.12528`.

[4] (2017). Corrigendum. [Correction to Avgar et al. (2016)]. *Methods in Ecology and Evolution*, 8(9), 1168. `https://doi.org/10.1111/2041-210X.12725`.

[5] A. Gupta, J. Johnson, L. Fei-Fei, S. Savarese and A. Alahi (2018). Social GAN: Socially Acceptable Trajectories with Generative Adversarial Networks. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2255–2264. `https://openaccess.thecvf.com/content_cvpr_2018/html/Gupta_Social_GAN_Socially_CVPR_2018_paper.html`.

[6] T. Salzmann, B. Ivanovic, P. Chakravarty and M. Pavone (2020). Trajectron++: Dynamically-Feasible Trajectory Forecasting with Heterogeneous Data. In *Computer Vision – ECCV 2020*, Lecture Notes in Computer Science, 12363, 683–700. Springer. `https://doi.org/10.1007/978-3-030-58523-5_40`.

[7] P. Kidger, J. Foster, X. Li and T. J. Lyons (2021). Neural SDEs as Infinite-Dimensional GANs. *Proceedings of the 38th International Conference on Machine Learning*, Proceedings of Machine Learning Research, 139, 5453–5463. `https://proceedings.mlr.press/v139/kidger21b.html`.

[8] T.-T. Gao, B. Barzel and G. Yan (2024). Learning interpretable dynamics of stochastic complex systems from experimental data. *Nature Communications*, 15, 6029. https://doi.org/10.1038/s41467-024-50378-x.

[9] T. Gneiting and A. E. Raftery (2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477), 359–378. https://doi.org/10.1198/016214506000001437.

[10] T. Gneiting, L. I. Stanberry, E. P. Grimit, L. Held and N. A. Johnson (2008). Assessing probabilistic forecasts of multivariate quantities, with an application to ensemble predictions of surface winds. *TEST*, 17, 211–235. https://doi.org/10.1007/s11749-008-0114-x.

[11] P. E. Kloeden and E. Platen (1992). *Numerical Solution of Stochastic Differential Equations*. Springer. https://doi.org/10.1007/978-3-662-12616-5.

[12] A. Nichol, J. Achiam and J. Schulman (2018). On First-Order Meta-Learning Algorithms. [Preprint], *arXiv:1803.02999v3*. https://arxiv.org/abs/1803.02999v3.

[13] S. Holm (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2), 65–70. https://www.jstor.org/stable/4615733.

[14] A. C. Davison and D. V. Hinkley (1997). *Bootstrap Methods and their Application*. Cambridge University Press. https://doi.org/10.1017/CBO9780511802843.

[15] NOAA Global Monitoring Division (n.d.). *General Solar Position Calculations*. [Online documentation]. Accessed 7 October 2026. https://gml.noaa.gov/grad/solcalc/solareqns.PDF.

[16] D. Zanaga et al. (2022). *ESA WorldCover 10 m 2021 v200*. [Dataset], version v200. https://doi.org/10.5281/zenodo.7254221.

[17] Overture Maps Foundation (n.d.). *Attribution and Licensing*. [Online documentation]. Accessed 6 October 2026. https://docs.overturemaps.org/attribution/.

[18] B. Lehner and G. Grill (2013). Global river hydrography and network routing: baseline data and new approaches to study the world's large river systems. *Hydrological Processes*, 27(15), 2171–2186. https://doi.org/10.1002/hyp.9740.

[19] HydroSHEDS (n.d.). *HydroRIVERS product documentation and licence*. [Online documentation], version v1.0. Accessed 6 October 2026. https://www.hydrosheds.org/products/hydrorivers.

[20] Mapzen / Tilezen (2017). *Terrain Tiles: Types of Terrain Tiles and Skadi format*. [Dataset distribution documentation]. Document revision e3d4351; accessed 7 October 2026. https://registry.opendata.aws/terrain-tiles/. https://github.com/tilezen/joerd/blob/e3d4351ee3e5be333e23f47cb500e9a7c310656a/docs/formats.md.

[21] Mapzen / Tilezen (2017). *Terrain Tiles: Attribution*. [Online documentation]. Document revision d8f587b; accessed 7 October 2026. https://github.com/tilezen/joerd/blob/d8f587b73d26e0a0c42cdccd9ae8c55de4197763/docs/attribution.md.

[22] Copernicus Programme / Sinergise (2021). *Copernicus Digital Elevation Model: GLO-30 Public*. [Dataset distribution documentation], 2021 release; accessed 7 October 2026. https://registry.opendata.aws/copernicus-dem/.

[23] Overture Maps Foundation (2026). *2026-08-19 release notes*. [Dataset release documentation]. Data version 2026-08-19.0; accessed 7 October 2026. https://docs.overturemaps.org/blog/2026/08/19/release-notes/.